KIET GROUP OF INSTITUTIONS · ASSIGNMENT QUESTION 4

# EMERGENCY PANIC ALARM PCB DESIGN

Design and Development Using KiCad

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Subject: PCB Design

Assignment: Question 4

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INTRODUCTION

# An Emergency Panic Alarm System

An Emergency Panic Alarm System is an electronic safety device designed to alert nearby people during an emergency.

- It uses a push button to activate the alarm.
- A buzzer produces an audible warning.
- A red LED provides a visual indication.
- The circuit operates using a 5V DC power supply.
- The system provides a simple and quick emergency alert.

**Objective:** To design a simple, low-cost and reliable emergency alarm PCB using KiCad.

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PROBLEM STATEMENT

# Delayed help in emergencies

In offices and workplaces, employees may need immediate assistance during unexpected emergencies.

Traditional methods of requesting help may cause delays.

## Proposed Solution:

- Install a panic button at an accessible location.
- Pressing the button activates the buzzer.
- The warning LED turns ON simultaneously.
- Nearby personnel can identify the emergency and respond quickly.

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COMPONENTS REQUIRED

# Components Required

| Component | Specification | Quantity |
|---|---|---|
| Push Button | Normally Open Tactile Switch | 1 |
| Buzzer | 5V Active Buzzer | 1 |
| LED | Red LED, 5mm | 1 |
| Resistor | 330Ω, ¼W | 1 |
| Capacitor C1 | 100µF, 16V | 1 |
| Capacitor C2 | 10µF, 16V | 1 |
| Connector | 2-Pin Terminal Block | 1 |

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WORKING PRINCIPLE

# Working Principle

1. The circuit receives a regulated 5V DC supply.
2. Initially, the push button remains open.
3. When an employee presses the push button, the circuit is activated.
4. Current flows through the LED and buzzer branches.
5. The LED glows and the buzzer produces an audible alarm.
6. When the button is released, the alarm stops.

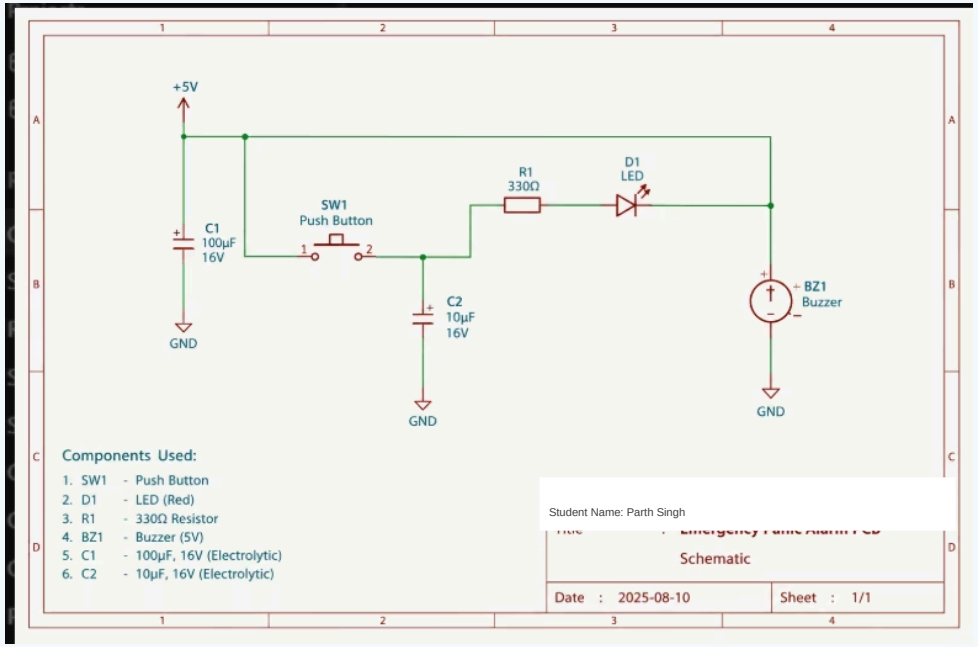
**Result:** Both visual and audible alerts are generated simultaneously.

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CIRCUIT SCHEMATIC

# Circuit Schematic

The circuit schematic represents the electrical connections between all electronic components.




## MAIN COMPONENTS AND THEIR FUNCTIONS

- **SW1:** Acts as the emergency activation switch.
- **R1:** Limits the current flowing through the LED.
- **D1:** Provides a visual warning indication.
- **BZ1:** Produces an audible warning.
- **C1:** 100µF capacitor for supply filtering.
- **C2:** 10µF capacitor for additional filtering.
- **J1:** Provides the 5V DC power input.

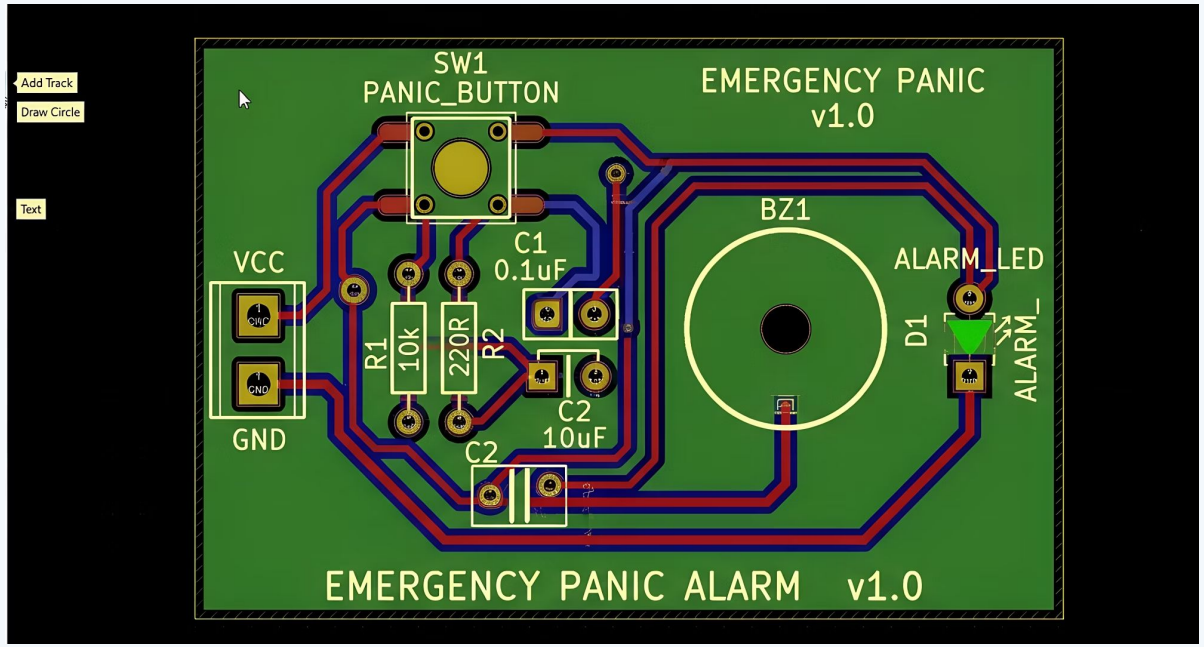
**Software Used:** KiCad Schematic

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FOOTPRINT ASSIGNMENT

# Footprint Assignment

A footprint represents the physical dimensions and pad arrangement of an electronic component on a PCB.




| Component | Suggested Footprint |
|---|---|
| SW1 | SW_PUSH_6mm |
| BZ1 | Buzzer_12x9.5mm |
| D1 | LED_D5.0mm |
| R1 | R_Axial_DIN0207 |
| C1 | CP_Radial_D6.3mm |
| C2 | CP_Radial_D5.0mm |
| J1 | TerminalBlock_2P_5.00mm |

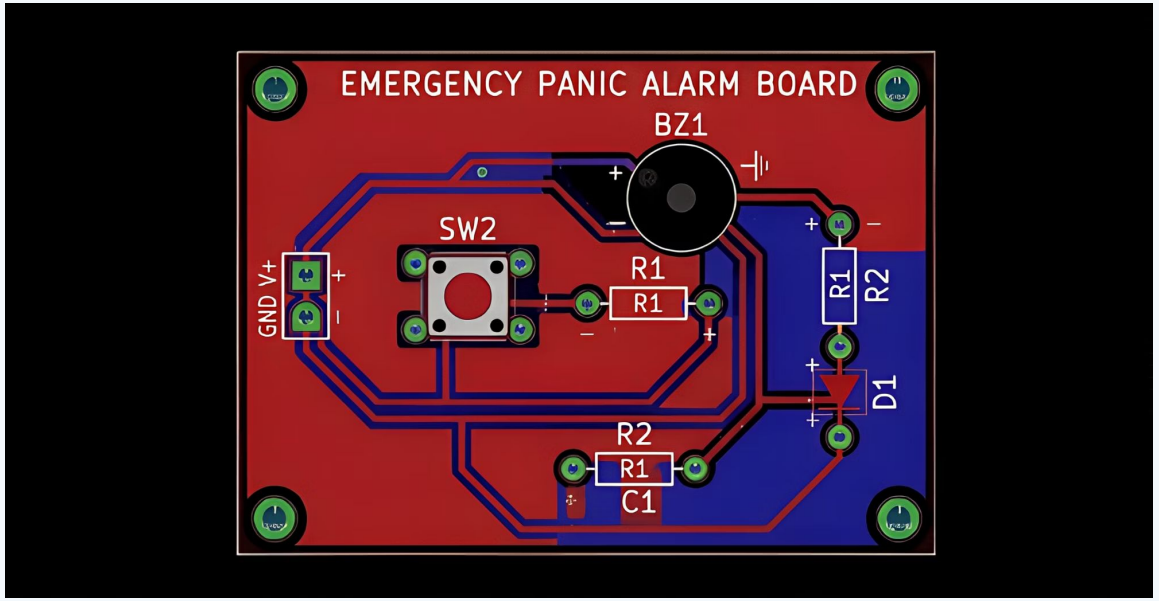
**Importance of Footprint Assignment:**

- Ensures correct component placement.
- Matches PCB pads with physical component leads.
- Helps avoid manufacturing errors.
- Improves assembly accuracy.

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# PCB Layout and Routing

PCB layout involves arranging electronic components and connecting them using copper tracks.




## DESIGN CONSIDERATIONS

- Place components according to circuit requirements.
- Keep connections short and organized.
- Maintain sufficient spacing between tracks.
- Use suitable track widths.
- Provide mounting holes where required.
- Verify electrical connections using the Design Rules Checker (DRC).

**Software Used:** KiCad PCB Editor

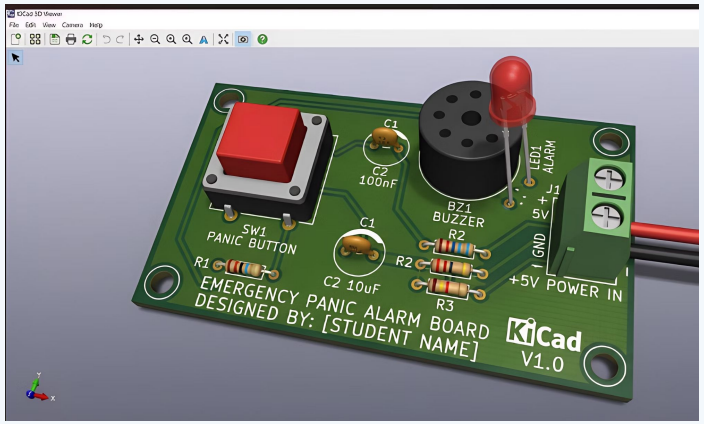
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# 2D and 3D PCB Views

## 2D PCB View

The 2D view displays the component footprints, pads, copper tracks and board boundaries.

It helps in checking the component arrangement and routing connections.




## 3D PCB View

The 3D viewer provides a realistic visualization of the PCB with mounted components.

It helps in understanding the physical appearance and placement of components before manufacturing.

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# Bill of Materials (BOM)

The Bill of Materials provides a complete list of components required to assemble the PCB.

| S.No. | Component | Reference | Quantity |
|---|---|---|---|
| 1 | Push Button | SW1 | 1 |
| 2 | Buzzer | BZ1 | 1 |
| 3 | Red LED | D1 | 1 |
| 4 | 330Ω Resistor | R1 | 1 |
| 5 | 100µF Capacitor | C1 | 1 |
| 6 | 10µF Capacitor | C2 | 1 |
| 7 | 2-Pin Connector | J1 | 1 |

**Total Components: 7**




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# Gerber File Generation

Gerber files are standard manufacturing files used to fabricate printed circuit boards.

## Steps for Generating Gerber Files:

1. Open the PCB Editor in KiCad.
2. Check the PCB layout and routing.
3. Open the Plot option.
4. Select the required copper and fabrication layers.
5. Generate the Gerber files.
6. Generate the drill files separately.
7. Save all manufacturing files in a dedicated folder.

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CONCLUSION

# Conclusion

The Emergency Panic Alarm PCB demonstrates the practical application of basic electronic components and PCB designing.

- A push button provides emergency activation.
- The buzzer generates an audible alert.
- The LED provides a visual warning.
- KiCad helps in schematic creation, footprint assignment and PCB layout.
- The design can be used as a basic emergency alert system.

## Future Improvements:

- Add a wireless notification system.
- Include a battery backup.
- Add a latching mechanism to keep the alarm active until reset.
- Add multiple panic buttons for different office locations.

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# THANK YOU

Emergency Panic Alarm PCB Design Using KiCad

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